SARWAN KUMAR PANDEY



AI/ML Engineer

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PROJECTS

Image Classifier

- Developed a CNN-based image classification web application using the CIFAR-10 dataset
- Built a FastAPI backend to serve real-time image predictions through a web interface
- Trained and deployed a TensorFlow/Keras model for multi-class image classification
- Integrated image preprocessing and inference using Pillow and NumPy

FaceID Studio – Real-Time Face Recognition

- Multi-threaded architecture — model loading, gallery building, and camera feed all run on background threads to keep the UI responsive
- Smart caching — gallery embeddings are cached with file-signature validation, so rebuilds only happen when images change
- Production-grade UX — settings panel for model selection, threshold tuning, and camera index; gallery manager with per-person image browsing and deletion

Retrieval-Augmented Generation (RAG) Pipeline

- Implemented a Retrieval-Augmented Generation pipeline for knowledge-based question answering
- Generated vector embeddings and performed semantic similarity search
- Improved response relevance by combining document retrieval with LLM-based generation

Food Review Sentiment Analysis

- Built a machine learning model to classify food reviews as positive or negative
- Performed text preprocessing and feature extraction using NLP techniques
- Evaluated classification performance using Accuracy, Precision, Recall, and F1-score

Image Classification using Convolutional Neural Networks (CNN)

- Built CNN models for image classification using TensorFlow/Keras
- Performed image preprocessing, model training, and evaluation
- Visualized model performance using accuracy and loss metrics

Sequence Prediction using Recurrent Neural Networks (RNN)

- Developed sequence prediction and time-series forecasting models using TensorFlow/Keras
- Implemented next-word prediction using Embedding and SimpleRNN layers
- Compared the performance of RNN, LSTM, and GRU architectures using Mean Squared Error (MSE)

SUMMARY

Aspiring AI/ML Engineer seeking an entry-level opportunity to apply my knowledge of machine learning, deep learning, and NLP to build real-world intelligent systems. Eager to contribute to innovative projects and grow as a software professional.

EDUCATION

B.Tech: Artificial Intelligence & Machine Learning

Dr. APJ Abdul Kalam TechnicalUniversity, Lucknow

2023 - Present

SKILLS

Machine Learning

Programming Languages

Python SQL C++ Java

CERTIFICATIONS

- **AI Fundamentals**
  IBM
- **Cloud Infrastructure AI Foundations**
  Oracle
- **GenAI Powered Data Analytics Job Simulation**
  TATA Forage
- **Software Engineering Job Simulation**
  JPMorgan Chase & Co
- **Machine Learning Foundations**
  AWS Academy Graduate
- **Generative AI Foundations**
  AWS Academy Graduate